

The neutrino feels the neutrons within $\lambda = 1/m_{Z_D}$ of it: one range, fixed by the mediator's mass, all along the way.
That matter is the cosmic mean for $\lambda \gtrsim$ Gpc, the Galaxy for kpc–Mpc, the Sun for $\sim$AU, the Earth for $\sim R_\oplus$, and the local density for $\lambda \ll R_\oplus$.

Distant sources

$\Phi_\nu \propto E^{-\eta}$, $z \lesssim 4$

Host halo

$(f_e : f_\mu : f_\tau)_S$

Cosmological propagation

Refraction
$V \propto g_D \epsilon \, u \, n_{\rm eff}$
$n_{\rm eff}$ smoothed over λ

$\nu + \bar{\nu}_{\rm relic} \to Z_D \to \nu\,\bar{\nu}$

Z_D

1 Gpc

Absorption + cascade

$E_{\rm res} \approx m_{Z_D}^2/2m_e$ $\quad m_{Z_D} \sim$ MeV

Milky Way

8 kpc

Sun

Adiabatic: populations kept

Galactic ν born in 3-D
Gas × cosmic rays; sources

Sun

$V \propto e^{-r/\lambda}/r$

1 AU

Coherent
Adiabatic ↔ sudden

Earth

Exterior

Mantle

Core

$R_\oplus$

IceCube: $\cos z = -\sin\delta$

Detection

HESE + through-going tracks

Cascade Track Double cascade

ν_e ν_τ ν_μ

Flavor

Sky maps

Extragalactic,
with z-evolution
Optional host galaxy

Refraction + absorption, one calculation
Thermal relic ν, Debye screening; cascade

$n_{\rm eff}$: Yukawa integral of
the Galaxy's neutrons
Galactic ν born in local eigenstates

Potential along paths
at all angles to the Sun
Its yearly motion averaged

Yukawa integral of
the Earth's neutrons
Exact chords, PREM

IceCube responses
MCEq + Barr
Flavor, spectrum, direction